

Kinematic parameter estimation technique for calibration and repeatability improvement of articulated arm coordinate measuring machines

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Received 24 May 2007; received in revised form 19 August 2007; accepted 5 September 2007

Available online 13 October 2007

Abstract

The kinematic modeling of articulated arm coordinate measuring machines (AACMM) has inherited both the previous developments in the field of robot arms and manipulators, and their calibration and parameter identification techniques, given the similarity of their mechanical characteristics. The different accuracy and repeatability requirements of both systems make it necessary to consider different identification techniques covering the characteristic operational parameters in each case. This paper presents a new data capture technique for subsequent identification of an AACMM kinematic model parameters, using nominal data reached by a ball bar gauge, along with the algorithm and objective functions used, based on a new approach including terms regarding measurement accuracy and repeatability. Thus, an estimation and error measurement correction method using a repeatability error model based on Fourier polynomials is derived from the identification scheme. A Sterling series FARO arm with 1.5 m long and 6 degrees of freedom (dof) was used to carry out experimental tests in order to evaluate the efficiency of the techniques presented, showing improved accuracy and repeatability performance.

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Keywords: Kinematic calibration; Coordinate measurement arms; Parameter identification; Data capture; Error correction; Repeatability

1. Introduction

The portable nature of the AACMMs, along with their level of accuracy, has favoured their use in inspection and quality control tasks in the industry, where the use of traditional coordinate measurement machines (CMMs) is not possible due to operational or cost reasons. In addition to contact measurement, the recent incorporation of non-contact measurement systems to AACMMs has broadened their sphere of action to reverse engineering and cloud of points inspection, leading to the boom experienced by the market over recent years.

AACMMs make up a special group within coordinate measurement due to their special characteristics and differences with regards to traditional CMMs. Although their function is the same, their fundamental difference is the kinematic structure. While the CMMs, whether they are bridge, gantry or horizontal arm type, have a cartesian configuration which allows the measurement of the physical displacement of each of the three linear

axes, in the AACMMs the measured point is a result of a series of coordinate transformations, depending on the model. AACMMs adopt the kinematic structure and model of robotic arms, and, similarly, are made up of a series of straight sections linked by rotary joints which provide them with the dof necessary to reach the required measurement positions. The differences compared to robotic arms are great, both in terms of accuracy and functionality, paying special attention to the accuracy of the sensors, materials used and dynamic conditions. Moreover, another important difference compared to CMMs and robotic arms is their manual and portable operation, instead of having machine axes automatically controlled.

The growing use of the AACMMs has been accompanied by an absence of standards on verification and calibration procedures, both from the point of view of the user and of the manufacturer. Traditionally, each AACMM manufacturer has adopted its own evaluation procedures. Firstly, it is necessary to determine the value of the parameters of the kinematic model of the arm. To this end, each manufacturer uses its own methods depending on the model and parameters implemented in each arm. Both the mathematical model considered and the method used to identify parameters constitute restricted information

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which is not available to the final user. This information is essential in order to deal with any accuracy evaluation of the manufacturer kinematic model, although it is not the main goal when establishing general evaluation methods.

These evaluation methods are based on the procedures set out by the predecessors of the three main standards for performance evaluation in current CMMs, UNE-EN ISO 10360, ASME B89.4.1 and VDI/VDE 2617. Evaluation tests based on these standards are still carried out today, in order to evaluate and compare the accuracy of an arm from the point of view of the CMMs. However, the special characteristics of the AACMMs require different verification procedures. While for a CMM an X , Y , Z point clearly defines a position of the three machine axes, for an AACMM the possible positions of its elements to achieve a fixed point defined in the measurement volume are practically infinite. Moreover, for CMMs, besides the general evaluation methods, it is possible to carry out evaluation tests in order to extract the positioning errors to finally implement correction models [1]. The CMM evaluation tests achieve a high level of maintenance of the physical–mathematical relations between the error model parameters and the error physically committed by the machine. This is not the case of AACMMs, in which the application of these models does not make sense, given the difficulty of directly relating the error committed with the model parameters, which are obtained by using optimization procedures.

The use of different methods based on mentioned standards and their adaptation to AACMMs results in a great heterogeneity in the presentation of results. In many cases this makes impossible a direct comparison between arms evaluated by different methods. In order to finish with this lack of normalization and standardize both the presentation of results and the methods used in evaluation, the only standard existing in the field of AACMM verification was recently developed. ASME B89.4.22-2004, which sets out the experience accumulated in definition and carrying out of evaluation methods of different manufacturers, recommends the methods which should be followed for reliable performance evaluation of the measurement arms. This document is an historical milestone in AACMMs evaluation, dealing with complete standardization in this field. It is focused on contact measuring with active and passive probes, covering the common measurement arm applications, specifically excluding evaluation tests for arms with optical sensors for non-contact measurement. In order to uniformize and eliminate ambiguity in evaluation methods for AACMMs, this standard fulfils its function, without making any indications with regards to parameter identification methods, calibration or correction.

The metrological behavior of an AACMM depends on many factors, although, basically, the final measurement error will depend on the correct determination of the mathematical model parameters [2]. The scant number of bibliographical resources in the field of AACMM modeling contrasts with the large number of studies regarding modeling and calibration of robots and manipulators, from which it is possible to extract conclusions in order to extrapolate their application in AACMMs. Since the development of the well-known model proposed by Denavit and Hartenberg (D–H) [3], multiple variants of kine-

matic models based on homogenous transformations have been considered. These later approaches solve problems of indetermination derived from the use of D–H model in robot calibration procedures. In [4], three levels of calibration for robots are established: joint level, kinematic model level and dynamic model level. The present work is focused on the second level of calibration. Thereby, the possible influences in the final measurement error will be studied to determine the mathematical model parameters. Those influences can be both geometrical – due to the kinematic model – and non-geometrical – due to other factors as assembly, damage in joints or transmission errors –. The approach of models which try to separate geometrical and non-geometrical errors has been a constant in literature on robot arms, without any apparent success from the point of view of a generalizable model regarding both robots and AACMMs. Based on the mathematical separation of both influences through the consideration of additional parameters in the model, the determination of a set of parameters related to each type of modeled error is carried out following optimization techniques without maintaining a direct relation between the real physical parameters and those obtained by optimization. Thus, these parameters cannot be considered more than an adjustment to the captured data which minimizes the positioning error in the case of robots but which do not have any numerical restriction to maintain the relation between results and real physical parameters, not being justifiable in many cases additional parameters that cause redundancy and complicate unnecessarily the mathematical model. In [5] the relationship between the mathematical results and the physical reality, along with the influence of the redundancy in model parameters on a calibration method, are studied. The detection of the error sources is carried out by the way the model is approached. However, traditionally, the second calibration level only studies the identification of geometric parameters, whereas the non-geometrical and temperature influences are considered in a different way from model to model.

Regarding the D–H model itself, Hayati and Mirmirani [6,7] outlined the discontinuity and non-proportionality of the model proposed by D–H for robotic arms with parallel or nearly parallel joint axes. In these cases the elimination of a parameter and the inclusion of a new rotation parameter are proposed. Many other authors have tried to solve the problem of the model proposed by D–H. In [8] a modification of the D–H parameterization is proposed, adding a fourth joint parameter which describes with a variable the reference frame position in the direction of the previous frame axis. Subsequently, new approximations considering the general equation of rigid body spatial movement [9] to model transformations between joints appeared, solving the indetermination of the D–H model but introducing redundant parameters. However, it is desirable, and indeed essential for many researchers, that a calibration model is free of redundancies in order to identify each error with a parameter, to establish reliable correction methods and even to improve the machine design. This assumption cannot be generalized, since to detect an error and to find its relation to a particular source depends on many other factors. Although [10] and [11] describe the maximum number of independent parameters necessary in order to define the geometry of an arm depending on the type

of joints, redundancy may exist. In [12], a series of considerations regarding the minimum number of parameters to describe the kinematics of an arm can be found, along with the need for redundancy in certain robot configurations in order to maintain the physical–mathematical link in model parameters.

The specific inclusion of non-geometrical errors effect on a robot calibration process by mean of model parameters or series approximation of them appears in [13] and [14], beyond considering the arm errors in terms of variation of geometrical parameters. In [15] there is an approximation to the non-geometrical errors model based on the mathematical model proposed by Sheth and Uicker [16], which, like the previous ones, allows maintaining the nominal positions of the joints reference frames, since it uses different frames to model the geometrical and non-geometrical errors. In this way, the definition of the nominal model does not require any particular modeling procedure, since the errors are modeled by way of new matrices and transformations to the nominal systems. Subsequently, the inclusion of intermediate error matrices has been given extensive consideration in the biographical resources on robots to model non-geometrical errors. In [17], an efficient method to consider any kind of errors is shown, including an intermediate matrix which relates joint nominal frame position with a second joint frame in its real position.

Stone et al. [18] proposed a model, known as *S-model*, which adds two parameters to the basic D–H notation in order to allow variable positioning of the reference frames. Again, a solution to the indetermination problem is obtained, resulting in a complete but non-proportional model. Hollerbach and Wampler [19] later proposed a model based on a mix of the D–H parameters and the Hayati notation, using model parameters depending on the type of joint in order to avoid indeterminations. In the search for a model which fulfils the three basic properties to be used for calibration [10], Zhuang and Roth [20] proposed the CPC model. Based on the idea that an incomplete model can be made complete by adding parameters, from the D–H model, they considered two more parameters to model the arbitrary position of the frames. Finally, models which consider the physical reference position of the encoder zero and which determine the relation between this position and the initial nominal position of the model [9,13,21] should be considered as the last main group of kinematic models.

Apart from the modeling techniques presented, all of them with later approximations in bibliography, and those which combine parameters associated to geometrical and non-geometrical error [22,23] – given the difficulty of modeling separately –, one of the current modeling trends involves an approximation to the final error based on the simplest kinematic models [24]. In this way, they do not try to explicitly model each parameter based on a parametric model which includes the error. So far, the models reviewed increase in mathematical complexity when modeling each supposedly identifiable error by way of parameters not directly measurable. Once the value of these parameters has been established by optimization, it is possible to correct the final position of the end of the arm through an error model using off-line correction tools. Using different linear and non-linear regression mathematical tools, such as polyno-

mials, Bezier curves, Fourier series, wavelets, neural networks, etc., it is possible to make an approximation to the arm error in line with the error characteristics observed in different positions of the work space. The non-parametric nature of these tools makes more difficult to link the geometrical and non-geometrical parameters of a model with the physical reality of the arm. Most applications which use regression as an approximation to errors try to combine traditional modeling and regression techniques to allow the error to be linked in some way to measurable arm parameters.

In this way, despite not fulfilling the conditions of proportionality and equivalence [10], the D–H model avoids redundancies and perfectly describes the kinematics of a measurement arm, not presenting indeterminations in practically all AACMMs on the market, whose dual joints define consecutive perpendicular axes. For this reason, in the present work the AACMM has been modeled by way of D–H when dealing with the developed identification procedure, although it is easily generalizable to any kinematic model. Software integration of the developed identification and error correction techniques for the experimental tests carried out is the first step towards the implementation of free software applied to coordinate measurement with AACMMs. This will establish the basis to, once integrated encoder drivers, obtain in an efficient way the measured point coordinates and probe orientation for any kinematic model considered.

2. Kinematic model

The static calibration of an AACMM establish a parametric model of its kinematic behavior in order to determine, numerically, the relationship between the joint variables and the probe position for any arm posture. A direct kinematic model takes the form of Eq. (1).

$$y = f(\theta_i, q) \quad (1)$$

with $i = 1, \dots, n$ for an arm with n rotating joints. This model calculates the position and orientation of the AACMM probe y , according to the value of the joint variables θ_i and to the equations of the model defined in f , which depend on the parameters vector q . This parameters vector contains the geometric parameters of the model, which must be optimized in order to obtain the lowest possible measurement error. Depending on the chosen kinematic model, the way the equations are obtained in f changes, along with the number of geometric parameters necessary to be included in q . The D–H basic model uses four parameters (d_i , a_i , θ_i and α_i) to model the transformation of coordinates between successive reference systems. The homogeneous transformation matrix between frame i and $i - 1$ (Eq. (2)) depends on those four parameters.

$${}^{i-1}A_i = T_{z,d}T_{z,\theta}T_{x,a}T_{x,\alpha}$$

$$= \begin{bmatrix} \cos \theta_i & -\cos \alpha_i \sin \theta_i & \sin \alpha_i \sin \theta_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \alpha_i \cos \theta_i & -\sin \alpha_i \cos \theta_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Calculating successive transformations of coordinates, by pre-multiplying the transformation matrix between a frame and the previous one, it is possible to obtain the global transformation matrix of the arm, which obtains coordinates of the center of the probe sphere with respect to the base of the AACMM.

$${}^0T_6 = {}^0A_1 {}^1A_2 {}^2A_3 {}^3A_4 {}^4A_5 {}^5A_6, \quad \bar{X}_{\text{AACMM}} = {}^0T_6 \bar{X}_{\text{Probe}} \quad (3)$$

In this manner, considering 0 as the global fixed reference system of the base and reference frame 6 moving with the rotation of the last joint, the desired homogeneous transformation can be obtained by way of Eq. (3). Apart from the four joint parameters indicated, it is necessary to consider an extra joint parameter θ_{0i} which will define the relationship between the encoder physical reference mark position of each joint and the initial position considered in the definition of the kinematic model. Thus, the joint variable θ_i of the model is related to the rotation reading provided by the encoder through Eq. (4), where θ_{0i} must also be identified from its nominal value defined for the initial position.

$$\theta_i = \theta_{i\text{Enc}} - \theta_{0i} \quad (4)$$

There are different alternatives to define the positioning of the hand frame in the case of robotic arms. The use of a reference system defined in the robotic hand responds generally to the need to have a frame which follows the characteristic directions of the tool, which will be subsequently controlled. In this case, since the aim is to obtain the sphere center coordinates of a static probe, the use of a reference system attached to the probe is omitted, assuming the link between the probe center and frame 6 to be a translation expressed by the coordinates X_{Probe} , Y_{Probe} , Z_{Probe} of

Table 1

Initial values for D–H parameters at AACMM modelling pose (Fig. 1)

Joint	a_i (mm)	α_i (°)	d_i (mm)	θ_{0i} (°)
1	0.0	−90.0	0.0	0.0
2	0.0	90.0	47.8	15.0
3	0.0	−90.0	645.83	−90.0
4	0.0	90.0	54.5	0.0
5	0.0	90.0	615.7	90.0
6	0.0	0.0	0.0	0.0

X_{probe} (mm): 0.0; Y_{probe} (mm): 138.5; Z_{probe} (mm): 54.5.

the sphere center respect to this frame. With this, the model has 4 joint parameters plus the three coordinates of the probe sphere center in frame 6, what makes a total of 27 parameters for the 6 dof AACMM considered. The initial values taken for the arm model parameters, obtained from CMM measurements and for the initial position, are shown in Table 1.

3. Parameter identification

The AACMM used in the present work is a 6 dof Sterling series FARO arm with a typical 2-2-2 configuration and a-b-c-d-e-f deg rotation, with a nominal value of $2\sigma = \pm 0.102$ mm obtained in a single-point articulation performance test of the arm manufacturer, without specifying the number of positions of the kinematic seat and points captured, and currently out of calibration. Firstly, simple evaluation tests have been carried out to evaluate the current state of the measurement arm using the manufacturer capture software and model, in order to know the expected initial values for accuracy and repeatability with

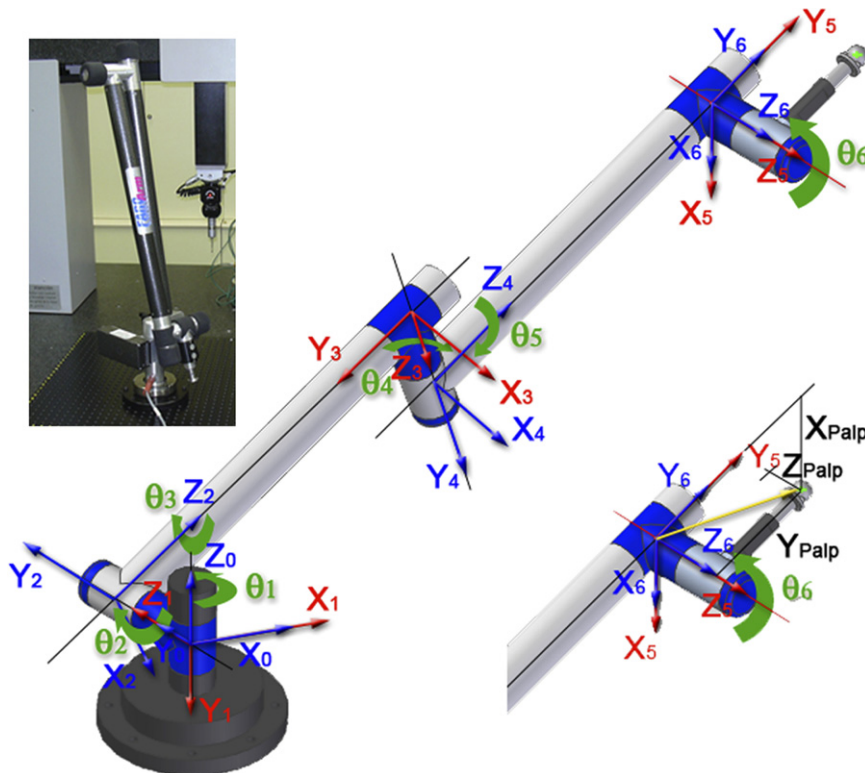


Fig. 1. Model definition posture of FARO AACMM with D–H convention. Initial values of parameters are shown in Table 1.

the arm in the current situation. Using a spherical-point passive probe of 6 mm in diameter, 14 spheres of a ball-bar gauge have been measured in a single gauge position, obtaining a mean deviation of the lengths measured with regards to the gauge lengths of 0.854 mm. On the other hand, a single-point articulation performance test has been carried out using a conical seat to capture 100 points in different AACMM orientations for a single kinematic mount position. The result obtained in this test was 0.347 mm for 2σ , with respect to the mean point of the captured data. Without the need for further initial evaluation tests or complete tests in accordance with ASME B89.4.22-2004, it can clearly be appreciated that the AACMM is out of calibration with the manufacturer parameters and model. Hence, it will be taken as the initial situation when dealing with the proposed calibration procedure.

3.1. Data capture setup

Once the mathematical model is defined, the next step involves the capture of nominal coordinates in the workspace of the AACMM. All the calibration procedures, both for robotic arms and AACMMs, establish a system which acquires coordinates or nominal distances in the workspace, in order to capture points which allow the error to be evaluated and minimized. The amount of identification data capture methods for robotic arms contrasts with the scant bibliographical resources regarding capture methods for parameter identification in measurement arms. Kovac and Frank [25] propose the use of a high precision gauge instrument for verification and calibration of AACMMs. Their work is centered on the development, manufacture and characterization of this instrument for use in parameter identification procedures and evaluation tests for AACMMs in laboratories, determining the gauge uncertainty due to its main influence variables.

Current identification procedures for robotic arms are based on a small set of positions where the coordinates of the robot hand are captured discretely. In [24], based on the results obtained in [26,27], 85 discrete robot positions were selected to cover the entire work range and extrapolate the results obtained to any arm position. The selection of capture positions to identify kinematic parameters must be preceded by tests which characterize the influence of each joint on the final error, to finally choose positions in accordance with this influence. For this reason, the application of a number of specific positions is not generalizable from one robotic arm to another, since the errors committed by each robot will depend on their configuration and assembly defects. In [28], the method described to identify the geometrical parameters takes 30 capture positions, without any apparent justification. It is common to consider a very different number and type of robot positions in the bibliography, constituting a very important step of the procedure for the extrapolation of the results. Recent research [29] deals with statistical methods to select the minimum number of necessary positions in order to obtain reliable results following optimization. The definition of positions for identification in robots aims to cover the maximum range of joint rotation, considering that, in this way, the influences of all the robot arm elements in the workspace are

Table 2

Ball bar positions in quadrant 1 in data capture setup for parameter identification

Pos. no.	Affected quadrants	Inclination	Distance B1	Direction
P1	Q1-Q2-Q4	Horizontal	Medium	Tangential
P2	Q1-Q4	Horizontal	Near	Tangential
P3	Q1-Q2	Horizontal	Far	Tangential
P4	Q1	Horizontal	Near	Radial
P5	Q1-Q2-Q4	+45°	Medium	Tangential
P6	Q1-Q2-Q4	−45°	Medium	Tangential
P7	Q1	Vertical	Medium	Tangential

covered. The referenced procedures successfully carry out their task, and hence both the capture of points and the subsequent test positions are captured discretely in robot configurations which are very similar to each other and also similar to the common robot work positions. This means that the positioning accuracy of the robot is improved globally in common work configurations. Since the optimization of the kinematic parameters is a least squares method, the adjustment of the parameters which minimize the error in the robot identification positions will mean small errors occur in positions similar to those of capture, but will cause bigger errors in very different positions.

In this work, a continuous data capture method has been developed. This technique allows the massive capture of arm positions corresponding to several points of the workspace. To this end, a ball-bar gauge of 1.5 m long was placed in 7 positions within the workspace of the arm in order to cover the maximum number of possible AACMM positions, to subsequently extrapolate the results obtained throughout the volume. Fig. 2 and Table 2 show the considered positions for the bar in a quadrant of the workspace. The ball-bar comprises a carbon fiber profile and 15 ceramic spheres of 22 mm in diameter, reaching calibrated distances between the centers with an uncertainty, in accordance with its calibration certificate, of $(1 + 0.001L)\mu\text{m}$, with L in mm.

The capture of data both for calibration and for verification of the arms is usually performed by way of discrete contact probing of surface points of the gauge in order to obtain the center of the spheres from several surface measurements. This means that the time required for the capture of positions is high, and then, identification is generally carried out with a relatively low number of arm positions. In the present work, a specific probe, capable of directly probing the center of the spheres of the gauge without having to probe surface points, was designed. The design of the probe is based on general considerations regarding the design of kinematic sphere mounts introduced in ASME B89.4.22-2004 appendix G. As seen in Fig. 3, the probe comprises three tungsten carbide spheres of 6 mm in diameter, laid out at 120° on the end of the probe. Since the ceramic spheres of the gauge have a diameter of 22 mm, it is necessary to establish the geometrical relationships in order to ensure the proper contact of the three spheres and the stability of this contact. In general, in order to maintain this stability, it is recommended a contact between the spheres of the kinematic mount and the sphere to fit between them at 45° with respect to the plane formed by the centers of the mount spheres. Thereby, the centering of the probe direction

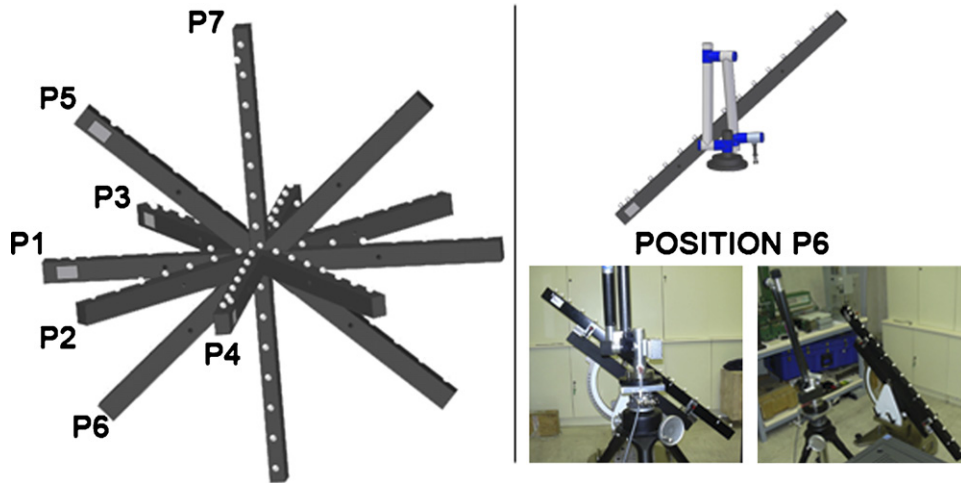


Fig. 2. Ball bar locations for each quadrant. Sample of position P6.

with regards to the sphere center is ensured, making this direction cross it (Fig. 3) for any orientation of the probe. Thus, it is possible to define a probe with zero probe sphere radius and with the distance from the position of the housing to the center of the probed sphere of 22 mm as length, allowing direct probing of the sphere center when the three spheres of the probe and the sphere of the gauge are in contact. This configuration also allows us to measure the initial value of the three last parameters of the model X_{Probe} , Y_{Probe} , Z_{Probe} , to define the coordinates of the center of the probed sphere with regards to the last measurement arm reference system. Moreover, this probe will be considered as the reference probe for the later kinematic parameter identification. The main reason to introduce these three parameters in the optimization of the model is to correct the deviations pro-

duced in the probe distances with respect to the last frame, both for imperfections in the assembly of the kinematic mount and for variation of the position of the last arm frame after optimization.

Besides characterizing and optimizing the behavior of the arm with regards to error in distances, its capacity to repeat measurements of a same point is also tested. Hence, an automatic arm position capture software has been developed to probe each considered sphere of the gauge and to replicate the arm behavior in the single-point articulation performance test, but in this case, to include the positions captured in the optimization from the point of view of this repeatability. The rotation angle values of the arm joints for each position, reached in the continuous probing of each sphere, are stored to obtain the coordinates of the measured point with respect to the global reference system

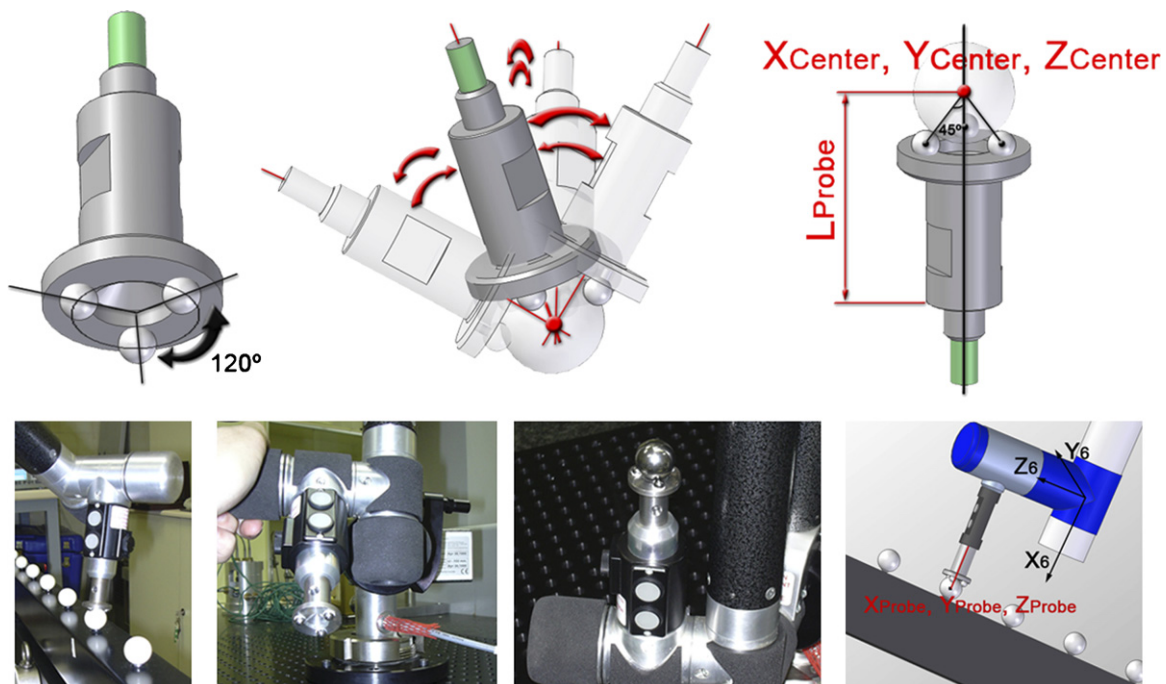


Fig. 3. Inverse kinematic coupling probe used in the capture of AACMM positions for parameter identification.

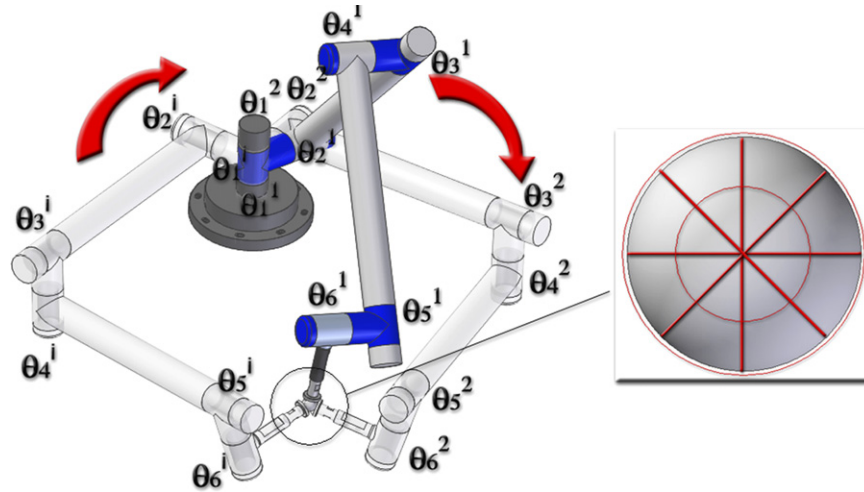


Fig. 4. Data capture procedure and capture trajectories. The readings from each of the 6 joint encoders are stored continuously for all capture AACMM positions.

for any set of parameters considered. In this way, with the kinematic mount probe in contact with the sphere, it is possible to capture the maximum possible number of arm positions, thus covering a large number of arm configurations for each sphere considered. Fig. 4 shows the capture scheme followed. As a general rule, the indicated trajectories will be followed for a probed sphere. Moreover, positions causing maximum variation of the arm joints in all the possible directions at the start, end and mid-point of each trajectory will be searched. The capture will be continuous and we will try to capture data in symmetrical trajectories in the sphere, in order to minimize the effect of probing force on the gauge. Thereby, around 400 rotation angle combinations θ_{iEnc} ($i = 1, \dots, 6$) have been captured for the joints to cover the positions of the arm probing the center of the measured sphere.

With this configuration, 4 spheres of the gauge in each of the 7 positions considered for each of the quadrants of the arm work volume were probed, what makes a total of 10,780 angle combinations. The measuring of a sphere center with the kinematic mount probe from different arm orientations should result in the same point measured. The unsuitable value of the kinematic parameters of the model will be shown by way of a probing error. This error produces different coordinates obtained for the same measured point in different arm orientations. In this manner, by probing four spheres of each position of the gauge with an approximate average of 400 arm positions per sphere, a series of 400 XYZ coordinates measured for each sphere center will be obtained. The deviations, initially due to the value of the parameters of the model between these 400 points in each sphere, will be used to characterize and optimize the arm point repeatability. In addition, in each gauge location 6 nominal distances between the four probed spheres are reached (Fig. 5a). The nominal distances of the gauge will be compared to the distances measured by the arm. Since an average of 400 centers per sphere are captured, the mean point of the set of points captured will be taken as the center of the sphere measured, in order to determine the distances between spheres probed by the arm (Fig. 5b). Thereby, a method for the subsequent combined optimization of the AACMM error in distances and point repeatability is defined.

3.2. Data capture evaluation

Due to the extreme importance of the number and type of data captured for identification, once the data are captured, the variation range for all the arm joints during capture has been analyzed. The optimization procedures based on least squares

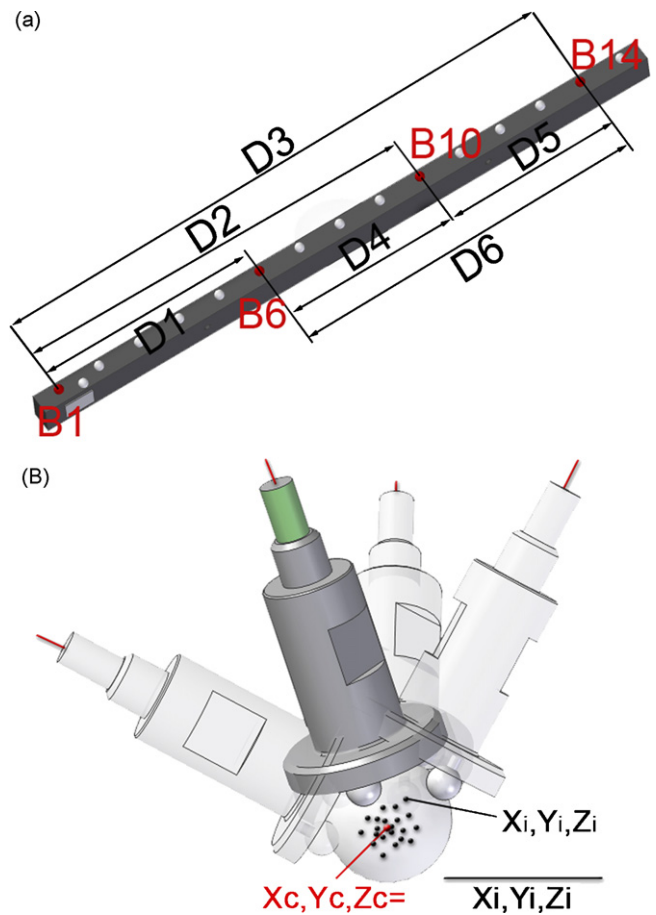


Fig. 5. Nominal parameters used in identification: (a) distances between spheres centers and (b) center considered to evaluate distances between spheres measured and point repeatability.

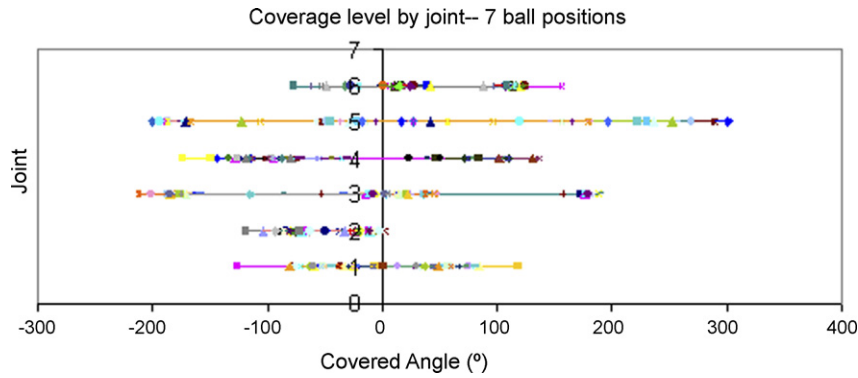


Fig. 6. Rotation range covered by each joint in complete data capture procedure for parameter identification (quadrant 1).

fitting will vary the values of the parameters in order to minimize the error obtained in the considered positions. Therefore, a low value of joint angle combinations with an initial value for parameters sufficiently close to the optimum will result in a low final error. Nevertheless, there is no guarantee that in positions outside of those used for optimization this error value will be maintained. As pointed in the previous section, the selection of the minimum number of arm positions necessary is a complex problem which, even when resulting effective, requires massive prior measurement of data for each tested arm, and is not generalizable. It is not possible to guarantee that a number and type of positions which are effective to identify the parameters of a type of arm model will be equally valid for another, where the influences on the final error may be originated in joint values different to those sampled by way of the same procedure based on the same capture positions. Hence, compared to the discrete position methods implemented in robots, a continuous capture procedure makes the sampling of the influences on the error more probable, without introducing discontinuities in the rotation ranges and without limiting the optimization to specific discrete combinations of joint rotation angles. Fig. 6 shows the rotation angles covered for each joint in the continuous capture of combinations for the 28 spheres probed.

The purpose of Fig. 6 is to check that there are no discontinuities between the rotation limit values covered by each joint, to ensure the capture of any possible influence on the final error. With this, it is possible to define the coverage level as a percentage of the joint rotation with regards to a complete rotation. Thus, it is possible to define a minimum coverage level for the data captured. As long as this minimum coverage level is reached, the results obtained by a subsequent optimization procedure can be considered as offering generalizable results to the arm measurement volume. In other words, the measurement error obtained in

any arm position shall be less than the maximum error obtained for the identification positions. Once the continuity in the capture of angle values between the detected end values has been checked, it is possible to define the level of global coverage obtained for all the arm positions measured. This level of coverage, obtained for the data captured in a quadrant of the arm, is shown in Table 3.

In the complete capture procedure, completing 100% of the coverage level in the joints in which it is possible should be aimed. It may be concluded that it is possible to select arm positions for optimization to obtain the maximum coverage level, and to eliminate ball bar positions which give rise to redundant angle regions. For instance, a coverage level similar to that of Table 3 is obtained only considering positions 2, 4 and 6 of the ball bar. However, the identification considering only these positions results in bigger error for positions different to those used in identification than considering the 7 measured positions. This shows the importance not only of a high coverage level, on which the selection of identification postures in [28] and in the majority of procedures reviewed in the bibliography are based, but also the combined effect of the joints on the error. In the case of robots, a continuous capture procedure involves the use of a high range tracking measurement instrument in order to obtain data for identification. Given the greater influence on their behavior of the inertia and dynamic loads during movement, and the fact that they generally work on static positions, continuous capture procedures are not as suitable for robots as they are for manual measurement equipment. For this reason, the use of discrete positions for parameter identification is generalized in robots. The high influence of the data captured on the final error, observed in the case of the arm, leads to consider the need to capture more positions for parameter estimation in robot arms, both in the selection of justified positions and in the

Table 3
Global coverage level

	A1 (°)	A2 (°)	A3 (°)	A4 (°)	A5 (°)	A6 (°)
Maximum	119.015	2.103	189.878	136.472	300.922	155.894
Minimum	−125.593	−118.371	−212.719	−174.191	−199.974	−76.931
Angle covered	244.608	120.474	402.597	310.663	500.896	232.825
Coverage level (%)	67.947	33.465	111.833	86.295	139.138	64.674

selection of apparently unjustified positions. All of this beyond obtaining sufficient data in line with the number of parameters of the model to be identified and avoiding redundant parameters, observed in most of the referenced bibliography.

In order to analyze the metrological characteristics of the AACMM for a specific set of parameters, both the error in distances of the arm and the dispersion of the points captured for each probed sphere center will be studied. As can be seen in Fig. 5, the parameters to evaluate are the six distances between the centers of the four spheres probed by bar location and the standard deviation of the points captured for each of the spheres probed. The 3D distance between pairs of spheres, based on the mean points calculated in each of them, is shown in Eq. (5).

$$D_{ijk} = \sqrt{(\bar{X}_{ij} - \bar{X}_{ik})^2 + (\bar{Y}_{ij} - \bar{Y}_{ik})^2 + (\bar{Z}_{ij} - \bar{Z}_{ik})^2} \quad (5)$$

in which D_{ijk} represents the Euclidean distance between sphere j and sphere k of the gauge i location, with coordinates corresponding to the mean of the points captured for sphere j and sphere k (Eq. (6)).

$$\bar{X}_{ij} = \frac{\sum_{m=1}^{n_{ij}} X(m)_{ij}}{n_{ij}} \quad (6)$$

In Eq. (6), n_{ij} is the number of angle combinations captured for sphere j in identification position i of the gauge, analogously for the coordinates Y and Z . In this manner, considering D_{0jk} as the nominal distance between spheres j and k obtained in the gauge calibration table, it is possible to calculate the error in distance between spheres j and k in location i in accordance with Eq. (7).

$$E_{ijk} = \sqrt{(D_{ijk} - D_{0jk})^2} \quad (7)$$

Since there are 4 spheres per gauge location, a total of 6 distances in each one are calculated, bringing a total of 42 distances. Considering in the previous equations $i = 1, \dots, 7$, and j and k covering each one of the spheres of the gauge for each position (1, 6, 10 and 14), eliminating the terms in which $j = k$ comes about, and considering that $D_{ijk} = D_{ikj}$, it is possible to evaluate the errors in distances obtained for all the gauge locations. Moreover, as a second quality value of a set of parameters, the maximum standard deviation of the points captured in each of the measured spheres is chosen (Eq. (8)).

$$2\sigma_{Xij} = 2\sqrt{\frac{\sum_{m=1}^{n_{ij}} (X(m)_{ij} - \bar{X}_{ij})^2}{n_{ij} - 1}} \quad (8)$$

In Eq. (8), σ_{Xij} represents the standard deviation in coordinate X of the points obtained for sphere j of position i . Analogously for the coordinates Y and Z . Following the diagram presented in Fig. 7 we obtain the values of all the possible standard deviations and distances for the data captured, considering a given set of parameters. In accordance with Fig. 7, the quality indicators of the initial set of parameters considered in Table 1 are shown in Table 4. Table 4 shows, in the first column, the maximum error in distances for all the positions of the gauge, the position of identification in which it is produced and the distance at which

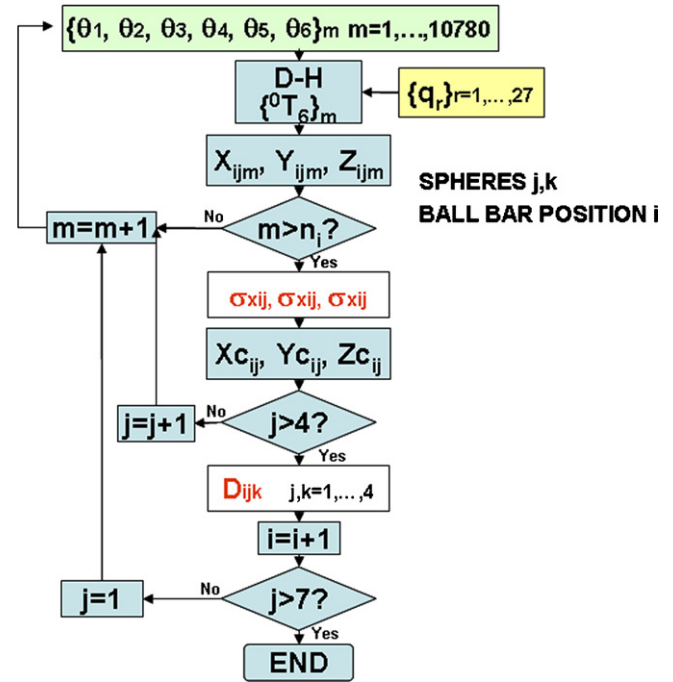


Fig. 7. Evaluation of a set of parameters q in identification positions.

the maximum error has been obtained. Likewise, the minimum error and its location, and the mean value of all the errors in distance observed are shown. In the case of standard deviation, Table 4 also includes the coordinate in which the value has been obtained, since both parameters are calculated separately for the three point coordinates. As can be seen, the values obtained for the initial set of parameters are large, as was expected given the initial lack of adjustment of the AACMM kinematic parameters.

3.3. Non-linear least-squares identification scheme

Kovac and Klein present in [30] an identification method based on nominal data obtained with the gauge developed in [25]. This method uses an objective function as used in robots, along with commercial software to identify kinematic parameters, without focusing the study on the particularities of the measurement arms. In [31], Furutani et al. describe an identification procedure for measurement arms and make an approximation to the problem of determination of AACMM uncertainty. This study is centered on the type of gauge to be used

Table 4

Quality indicators for the initial values of model parameters over 7 ball bar locations (10,780 AACMM positions)

Distance error (mm)		2σ by sphere (mm)	
Maximum	49,426767	Maximum	28,226716
Causing pos.	POS2	Causing pos.	POS 4
Causing dist.	D2	Causing sph.	B10
Minimum	0,275856	Causing coord.	Y
Causing pos.	POS2	Minimum	2,580066
Causing dist.	D5	Causing pos.	POS6
Medium	22,428725	Causing sph.	B10
		Causing coord.	X
		Medium	11,689359

according to the arm configuration and analyses the minimum number of necessary measurement positions for identification, as well as the possible gauge configurations to be used. Again, this work does not specify the procedure to obtain the parameters of the model, nor the type of model implemented, and does not show experimental results for the method proposed. In [32], Ye et al. develop a simple parameters identification procedure based on arm positions captured for a specific point of the space.

As indicated in Section 2, the kinematic model implemented in the measurement arm can be described, for any arm position, by way of Eq. (9), based on the formulation of direct kinematic problem.

$$p = f(a_i, \alpha_i, d_i, \theta_{0i}, X_{\text{Probe}}, Y_{\text{Probe}}, Z_{\text{Probe}}, \theta_{i\text{Enc}}) \\ i = 1, \dots, 6 \quad (9)$$

in which $p = [X \ Y \ Z \ 1]^T$ are the coordinates of the point measured with respect to the arm global reference frame at the base, corresponding to the value of the geometrical parameters and to the joints rotation angles in the current arm position. There are many alternatives when dealing with an optimization procedure, although the most widely used in the field of robot arms and AACMMs are the formulations based on least squares fitting. Given the non-linear nature of the arm kinematic model, it is not possible to obtain an analytical solution to the problem of parameter identification. Therefore, it is necessary to use non-linear optimization iterative procedures. In this way, for the mathematical formulation of the optimization method it is common to define the objective function to minimize in terms of square error components. Based on the nominal coordinates reached by the gauge and those corresponding to the points measured, we can obtain the arm measurement error as the Euclidean distance between both points, as shown in Eq. (5), although applied to the difference between the measured point and the nominal point. Since the identification procedure both in robots and in AACMMs is based on the capture of discrete positions within the workspace, all the reviewed optimization procedures use Eq. (10) as basic objective function to minimize.

$$\phi = \sum_{i=1}^m [\Delta p]^T [\Delta p] \\ [\Delta p]_i = [\delta x \ \delta y \ \delta z]_i^T = (p - p_0)_i \quad (10)$$

Eq. (10) quantifies the error in distances between the nominal point and the point reached for all the positions captured, formulated as the quadratic sum. There are variants of this expression in those cases in which a capture procedure based on nominal distances is proposed, where the measurement error of the arm is obtained in accordance with Eq. (7) for each distance considered, to obtain the errors in all the distances and with the target function being the quadratic sum.

In this work, in order to choose the objective function to be minimized, consideration has been given to the error in distances presented in Eq. (7) for the 42 distances measured. Therefore, it is possible to evaluate the 10,780 combinations of six values of joint angles for each set of kinematic parameters, and to obtain the centers as the mean value of the coordinates corresponding

to each sphere as shown in Eq. (6). Finally, we evaluate all the distances in each iteration of the optimization procedure. The objective function can be formulated as the quadratic sum of all the errors in distances calculated by way of Eq. (7). Hence an objective function similar to those commonly chosen in robot and AACMMs parameter identification is obtained.

Given the arm positions capture setup used, and the fact that point repeatability in any arm probe orientation is a very important parameter in order to characterize the metrological behavior, we use a new objective function in the optimization. Unlike traditional expressions, our objective function (Eq. (11)) includes both the errors in distance and the deviation of the points measured in each sphere showing the influence of the volumetric accuracy and point repeatability, minimizing simultaneously the errors corresponding to both parameters.

$$\phi = \sum_{i=1}^r \sum_{j,k=1}^s \left[(D_{ijk} - D_{0jk})^2 + (2\sigma_{xij})^2 + (2\sigma_{yij})^2 + (2\sigma_{zij})^2 \right] \quad (11)$$

In the objective function proposed, with the capture setup described, $r=7$ positions of the ball bar and $s=4$ spheres (1, 6, 10 and 14) per bar position. Again, in Eq. (11) it is necessary to consider the elimination of the terms in which $j=k$, in order to avoid the inclusion of null terms or considering as duplicate the influence of the error on distances, taking into account that $D_{ijk} = D_{ikj}$. The first term of Eq. (11) corresponds to the error in distances in position i of the gauge between sphere j and sphere k , whereas the other terms refer to twice the standard deviation in each of the three coordinates for sphere j in position i of the gauge. Finally, again by mathematical formulation of the optimization problem, it is necessary to consider the sum of all the square errors calculated. With the objective function of Eq. (11), 126 quadratic error terms will be obtained to calculate the final value of the objective function after each optimization algorithm stage. This value will show the influence of the kinematic parameters as well as of the joint variables through the calculation of the points coordinates corresponding to the 10,780 arm positions.

On the other hand, it is also possible to consider alternatives to the objective function of Eq. (11) in order to capture the influence of AACMM repeatability, replacing the standard deviation corresponding to the range or semi-range of coordinates calculated for each sphere, as shown in the objective function of Eq. (12).

$$\phi = \sum_{i=1}^r \sum_{j,k=1}^s \left[(D_{ijk} - D_{0jk})^2 + (\Delta x_{ij})^2 + (\Delta y_{ij})^2 + (\Delta z_{ij})^2 \right] \quad (12)$$

In Eq. (12), Δ represents the range in the corresponding coordinate for points of the sphere j in position i of the bar. The use of this type of objective function requires previous data filtering due to sensitiveness to the existence of spurious points amongst the data to be fitted. The experience obtained in the use of both objective functions, considering the terms regarding repeatabil-

Table 5
Identified values for the model parameters by L–M algorithm

Joint	a_i (mm)	α_i (°)	d_i (mm)	θ_{0i} (°)
1	0.036962	−90.052249	−0.000002	−0.126434
2	0.102485	90.044751	47.891183	14.942165
3	0.097868	−90.020699	645.780523	−88.99688
4	−0.133079	90.068899	54.240741	−3.636896
5	0.057606	90.011014	615.242600	89.770488
6	0.367275	−0.522698	0.150712	−0.878373

X_{probe} (mm): 0.367276; Y_{probe} (mm): 139.450887; Z_{probe} (mm): 54.657060.

ity grouped together by root mean square or separately as shown in Eqs. (11) and (12), shows that Eq. (11) concludes with better results, even without the existence of spurious points.

The Levenberg–Marquardt (L–M) method [33,34] has been chosen as optimization algorithm for parameter identification, given its proven efficiency in robot parameter identification procedures [12,24]. The selection of a specific optimization procedure implies to avoid the influence of the mathematical method itself with regards to the data captured on the result. One of the most suitable methods to solve this problem is the L–M algorithm. Table 5 shows the AACMM kinematic model parameters finally identified, based on the initial values of Table 1 for the objective function of Eq. (11) and the 10,780 arm positions considered. The error values obtained for the identified set of parameters are shown in Table 6.

In order to study the influence of the inclusion of the standard deviation on the objective function, we have complete optimizations taking as function only the terms corresponding to the error in distances Eq. (13) for the 10,780 positions captured, as would correspond to a common objective function for parameter identification of robots.

$$\phi = \sum_{i=1}^r \sum_{j,k=1}^s [(D_{ijk} - D_{0jk})^2] \quad (13)$$

Compared to the maximum and mean error obtained in Table 6, in this case a maximum error of 15 μm was obtained and a mean error of 5 μm for the same arm positions. However, for the parameters identified with the objective function of Eq. (13), the maximum value obtained for 2σ is 1.8932 mm compared to 0.249 mm obtained using Eq. (11), and the mean value is 1.009 mm. As can be seen in the results, an optimiza-

Table 6
Quality indicators for the identified set of model parameters over 7 ball bar locations with Eq. (12) as objective function (10,780 AACMM positions)

Distance error (mm)		2σ by Sphere (mm)	
Maximum	0.144258	Maximum	0.249325
Causing Pos.	POS2	Causing Pos.	POS 1
Causing Dist.	D1	Causing Sph.	B1
Minimum	0.005550	Causing Coord.	Z
Causing Pos.	POS1	Minimum	0.035286
Causing Dist.	D2	Causing Pos.	POS4
Medium	0.066202	Causing Sph.	B6
		Causing Coord.	Y
		Medium	0.104355

tion equivalent to those commonly found in robots produces excellent results for errors in distance but inadequate results for range and standard deviation. Hence, to obtain a set of parameters which allows the arm to be repeatable in a point for any measurement orientation and not only in the orientation captured for optimization, it is necessary to consider the range or the standard deviation in objective function. There may exist cases of robot arms in which an optimization scheme without considering repeatability evaluation parameters is useful for work positions and orientations similar to those used in identification. However, in general for robots and always in the case of AACMM parameter identification, regarding the standard deviation results, the traditional objective functions should be completed with repeatability evaluation parameters, obtaining kinematical parameters that makes more reliable the generalization to the measurement volume of the error values obtained. In the results, 7 standard positions in one of the four measurement arm quadrants where considered, although the method described is easily generalizable to any number of considered spheres and gauge positions.

4. Repeatability error model

Besides the positioning or measurement errors due to kinematic parameters, which is in general the biggest source of error, there are other error sources which influence the final accuracy of the arm. Dynamic, thermal, electrical or joint compliance errors can be considered. The optimization carried out tries to numerically adjust the parameters in order to commit the smallest possible error, not being possible to isolate and characterize the influences of kinematic parameters on this final error. Absorbing the error sources means that is not possible to separate and optimize each of them, resulting in a set of parameters which minimizes all errors. There are many approximations to the problem of geometrical and non-geometrical parameter identification and to the determination of their influences on the final error separately [22,23]. This is a common consideration in numerous robot models which eventually do not obtain the real influence. This complicates unnecessarily the mathematical models and does not establish a clear difference with regards to an optimization procedure without physical limitations. Moreover, it also breaks the physical–mathematical link required for the determination of isolated influences on the final error.

Also considering that the optimizations obtain a suitable point repeatability value but not too low compared to the reduction of error in distances (2σ around 0.4 mm with maximum errors in distance of 0.144 mm), it is possible to consider the acceptance of the kinematic parameters obtained, taking them as optimum, and to model the behavior of the points deviation based on the arm position. In this manner, we obtain a correction mechanism separate from parameter optimization, which will not have subsequent influences on the error in distances shown for the optimized kinematic parameters. This will allow us to reduce the deviation value obtained. Given the nature of the identification procedure proposed, in which the mean values of all the points captured in each sphere in terms of error in distances of the objective function are optimized, the correction model

should only be applicable to the error of each point with regards to this optimized mean, so that no influence is introduced on the measurement accuracy in distances, directly attributable to the kinematic parameters obtained. Thereby, we can consider the difference between each point and the mean point of the corresponding sphere – value which causes the standard deviations obtained – as an error due to the other influences, remaining separate from the kinematic parameters optimization. Moreover, this error can be modeled as an influence dependent on the rotation value of the joints.

There are several works in the field of robots which try to predict the error to be committed in a specific trajectory [24]. These studies do not consider point repeatability, since they propose kinematic parameter identification procedures based on discrete positions in the workspace. The error models analyzed for robot arms are based on the idea that any function can be approximated through the use of fitting techniques, in which a polynomial with suitable terms is adjusted to the error results obtained. Thus, it is possible to determine the error committed by the arm in each one of its positions, and approximate its value by way of ordinary polynomials, Bessel functions, Fourier series, Laguerre–Hermite, Jacobi, etc., or by way of the application of regression techniques based on neural networks or genetic algorithms. Once the function for approximation to the error values has been selected, in general, the correction models reviewed carry out adjustment tests in order to select the correct order of the polynomials used and the final terms considered, eliminating negative influences on the final adjustment. Subsequently, the model obtained is validated on arm positions different to those captured to fit the error function. In this case, given the similarities to this type of techniques, a similar sequence is followed to obtain the developed correction model, but taking into account the differences between a robot arm and an AACMM.

The joints in an AACMM are rotary, meaning that, although it is difficult to determine the influence of each of them separately on the final error, it can be assumed that the isolated influence of each joint is periodical with the joint rotation. There is research on robot arms showing the possibility of modeling errors due to structural deformations [2], roll errors in the joints gear trains [35], or errors due to the influence of the kinematic parameters as functions dependent on the rotation values of the joints [36]. In this work, the cyclical influence of each AACMM joint on the point repeatability error is analyzed. An initial approximation of several correction and error models proposed for robots [24] involves using ordinary polynomials as approximation functions, depending on the value of the rotation angle of the joints in each robot position. Eq. (14) shows an ordinary polynomial for the modeling of error in coordinate X considering s harmonics.

$$f_{\varepsilon X}(\theta_i, a_{xi}, b_{xi}) = C_X + \sum_{j=1}^6 \sum_{k=1}^s (a_{Xjk} \cdot \theta_j^k) \quad (14)$$

These polynomials, as is to be expected, offer good approximation in the range of experimental data collected, but deviate a great deal from the error function outside of this range,

which is a typical characteristic in any approximation using ordinary polynomials. Since the aim is modeling the periodic influence of the rotation value of a joint, the approximation by way of ordinary polynomials is not suitable since it does not suitably reflect the regularity of the occurrence of the influence of each joint, particularly in joints with infinite rotation capacity.

In this case, considering the repeatability error values described above, we can obtain a function of 6 variables, one for each joint angle, and try to approximate this function by a Fourier series. This approximation obtains a sum of periodic signals, much more suited to the real behavior of the arm. Therefore, an extrapolation of this model outside the range of data collected may be more precise than the use of polynomials. Based on the optimum parameters of Table 5, it is possible to reconstruct all the points captured during the identification stage and represent the errors in X , Y , Z of each one of the 10,780 points, corresponding to the 7 positions of the gauge bar with regards to the mean obtained for each sphere. If this error is represented consecutively for each position captured in identification phase, a function will be obtained. This function shows the value of the error and the corresponding position of the arm, since the 6 values of the joints rotation angles, which define this position, will correspond to each error value (Fig. 8). The expressions of the repeatability error for sphere j of position i of the gauge, for point k captured in this sphere, are shown in Eq. (15).

$$\begin{aligned} \varepsilon_{Xijk}(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6) &= \bar{X}_{ij} - X_{ijk} \\ \varepsilon_{Yijk}(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6) &= \bar{Y}_{ij} - Y_{ijk} \\ \varepsilon_{Zijk}(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6) &= \bar{Z}_{ij} - Z_{ijk} \end{aligned} \quad (15)$$

In accordance with Eq. (15), the error dependent on current position k of the arm for sphere j of position i of the gauge will be given by the difference between the probed point for this position and the mean point obtained for all the points probed for sphere j of position i . The repeatability error of each point is also defined independently for each coordinate.

Fig. 8 represents the repeatability error values in each coordinate for all the points captured for identification, obtained by way of Eq. (15) and showing the location of this error, indicating the sphere Bi of the corresponding gauge position. This function can be approximated by a Fourier series with sum of sine and cosine terms for each of the joint angles with coefficients to be determined by applying a non-linear regression method. In this way, it will be possible to capture the periodic influence of each joint rotation on the final error. These regression methods obtain an approximate function f as shown in Eq. (16).

$$y_X = f_X(\theta_i, a_{xi}, b_{xi}) + \delta_X \quad (16)$$

Eq. (16) represents the repeatability error in coordinate X , being possible to similarly consider the expressions for Y and Z . y_X represents the repeatability error function and f_X is a dependent function of the 6 joint angles and of the coefficients considered for the Fourier series. δ_X represents the residual. With this, in order to define function f which will approximate the repeatability error values, an ordinary Fourier series is defined

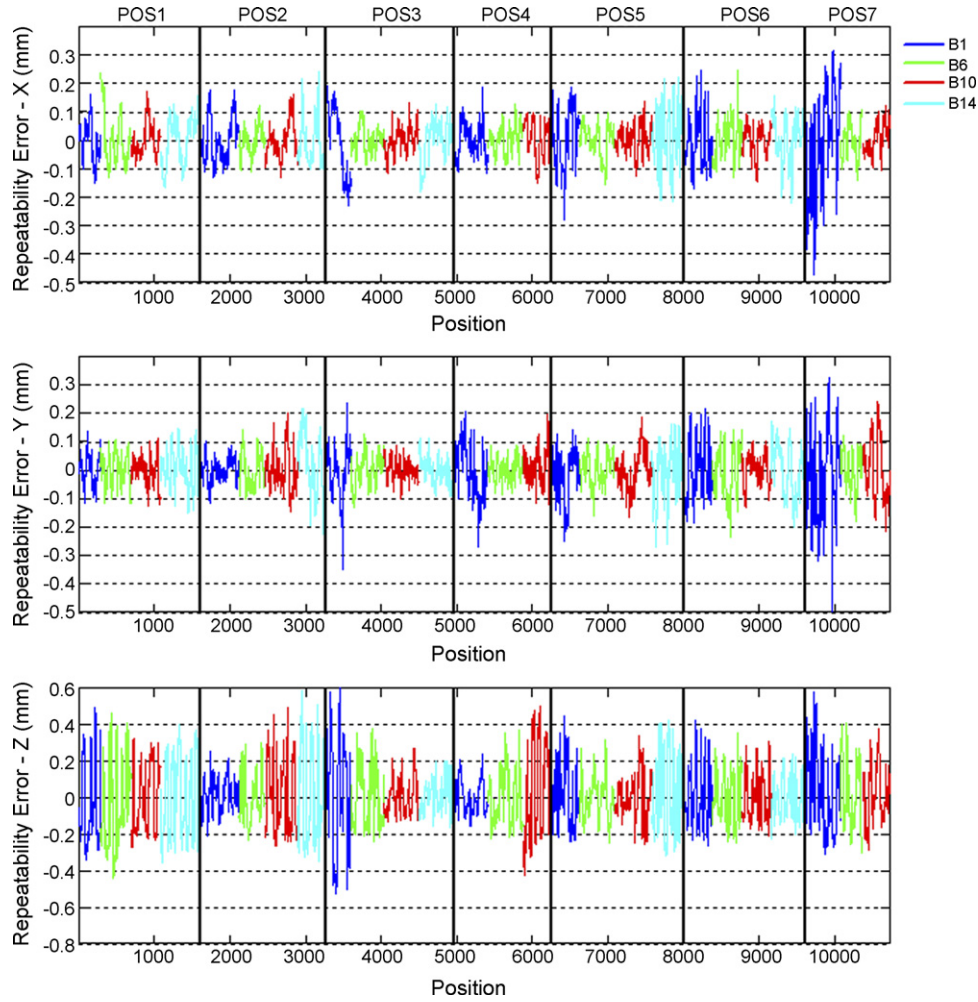


Fig. 8. Point repeatability errors (Eq. (16)) for the optimal set of model parameters over identification AACMM positions.

for each of the errors, corresponding to each coordinate (Eq. (17)).

$$\begin{aligned}
 f_{\varepsilon X}(\theta_i, a_{xi}, b_{xi}) &= C_X + \sum_{j=1}^6 \sum_{k=1}^t (a_{Xjk} \cdot \cos(k \cdot \theta_j) + b_{Xjk} \cdot \sin(k \cdot \theta_j)) \\
 f_{\varepsilon Y}(\theta_i, a_{yi}, b_{yi}) &= C_Y + \sum_{j=1}^6 \sum_{k=1}^t (a_{Yjk} \cdot \cos(k \cdot \theta_j) + b_{Yjk} \cdot \sin(k \cdot \theta_j)) \\
 f_{\varepsilon Z}(\theta_i, a_{zi}, b_{zi}) &= C_Z + \sum_{j=1}^6 \sum_{k=1}^t (a_{Zjk} \cdot \cos(k \cdot \theta_j) + b_{Zjk} \cdot \sin(k \cdot \theta_j))
 \end{aligned}
 \quad (17)$$

In Eq. (17), j is the articulation number and t is the number of harmonics to be considered within the Fourier series. With this approach, the value of the coefficients a and b for each angle and for each harmonic considered should be found, along with the coefficients of the cross products considered. To this end, given a set of repeatability error values known by way of Eq. (15) and combinations of 6 angles corresponding to each error value, we apply a least square non-linear optimization method to minimize the sum of squares of the differences between the real

and the approximate values by way of the functions proposed in Eq. (17). An approximation scheme based on the L–M algorithm has been used.

Regression tests have been carried out considering the fundamental harmonic ($t = 1, 2, 3$ and 4 harmonics). Values of the same order as the repeatability error were obtained for residuals for all the tests, improving significantly when increasing the number of harmonics. Therefore, it is necessary to increase the number of harmonics or to consider the influence of the cross products between terms. It should be remembered that adding more harmonics may be suitable to approximate the functions obtained with the data captured for the approximation model, although, in different measurement positions, it may be more effective to approximate only those components of low-frequency. These components, where they exist, will have an influence as a systematic error on the measurements. In this way, given the apparent random component in the approximated error functions, the search for the main error systematic components should be focused on the first harmonics of the Fourier series or by way of a PCA analysis. Moreover, the consideration of a greater number of harmonics may improve the approximation to the error in the identification positions considered, without

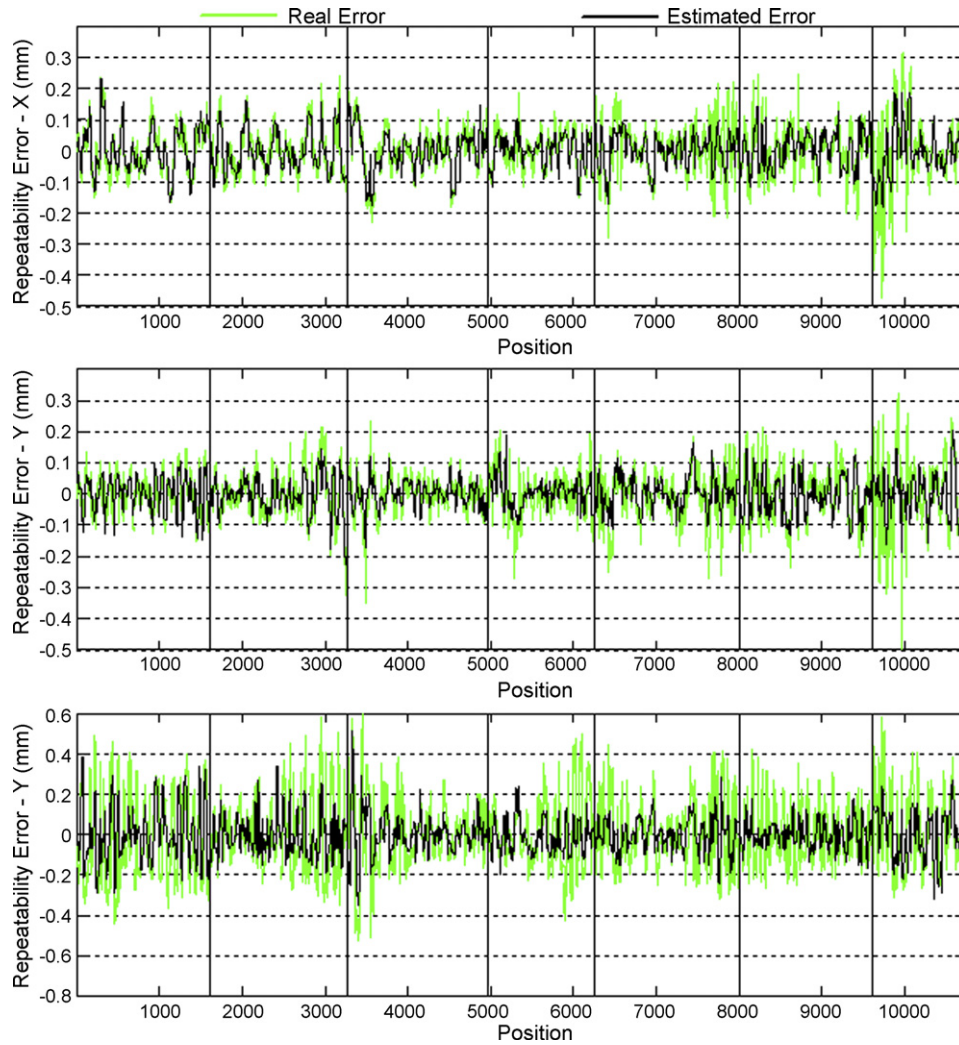


Fig. 9. Estimated and real repeatability error over identification positions through Eq. (17) with 2 harmonics and cross terms.

guaranteeing that it will also improve the error model in positions different to those of parameter identification. FFT of the repeatability error function shows a large number of frequencies, for which reason it may be more effective to consider the cross terms than increasing the number of harmonics in the approximation. The best result considering cross products in Eq. (17) is shown in Fig. 9. This result was obtained for Fourier series of 2 harmonics ($t=2$) plus the cross terms of the first harmonic, plus the cross terms of the second harmonic, plus the mixed cross terms, resulting in a dependent function of 219 parameters. The subsequent analysis of the reliability intervals of each of the parameters allows reducing the number of parameters necessary in the approximation.

5. Test and results

5.1. Kinematic model

The generalization of an identified set of parameters to the rest of the measurement volume involves the obtaining of devi-

ation and error values smaller than the maximums obtained for the identification process for any arm position. For this reason, the use of at least one test position different to the identification positions is recommended. Thereby, the maximum error for the identification positions, in those cases in which a lower number of gauge or arm positions have been taken, has proven to be better than that finally considered as optimum. However, in these conditions, the evaluation of the identified parameters on positions not considered before has resulted in worse values than those obtained in identification with consideration of all the positions captured. For this reason, the use of all the positions captured as representative of the arm measurement volume was the option taken. Thus, a sufficiently representative set is obtained in order to absorb all the influences on the final error and to obtain a set of kinematic parameters which make the error obtained in identification be realistic and truly the maximum for the arm for any position in the measurement volume. As is shown in Table 6, a maximum error of $144\text{ }\mu\text{m}$ and a mean error of $66\text{ }\mu\text{m}$ are obtained for all the measurement volume of quadrant 1. This can be compared to maximum

Table 7

Quality indicators for the identified set of model parameters over 14 spheres of ball bar test location (6,195 AACMM positions)

Distance error (mm)		2σ by Sphere (mm)	
Maximum	0.128411	Maximum	0.202460
Causing Pos.	POS.TEST	Causing Pos.	POS.TEST
Causing Dist.	D1-10	Causing Sph.	B14
Minimum	0.011107	Causing Coord.	Z
Causing Pos.	POS.TEST	Minimum	0.027287
Causing Dist.	D1-2	Causing Pos.	POS.TEST
Medium	0.068117	Causing Sph.	B3
		Causing Coord.	X
		Med. X	0.054182
		Med. Y	0.069676
		Med. Z	0.141153

error in distances (0.854 mm) and to mean error (0.262 mm) obtained in one single evaluation position in the initial situation. In normal operation of the arm – probing discrete points of the center of the sphere probe – the error obtained with the identified set of parameters will be normally around the mean value of 66 μm , producing the maximum error in certain specific arm positions.

Once the optimization process is complete, as the final stage of the presented parameter identification procedure, it is necessary to evaluate the behavior of the arm with the optimum set of parameters on arm positions different to those used during identification. The more similar the evaluation positions subsequent to those used in identification, the better the results. Hence, it is necessary to find different measurement arm positions to evaluate the level of fulfillment of the error values obtained in other measurement volume positions.

In this case, as test bar location subsequent to identification was chosen in the upper part of quadrant 1. Based on the same orientation of position P1, the bar was rotated approximately 25° both horizontally and vertically. For this ball bar location, angle combinations corresponding to the arm positions probing the centers of the 14 gauge spheres were captured. The kinematic coupling probe and the same capture conditions as those presented in the optimization procedure were used, that is, around 400 points per measurement sphere. In this way 6195 arm positions were captured for the test position, which is a reliable check of the measurement arm error on positions not used. Table 7 shows the error values obtained for the 14 test position spheres.

As can be seen in the results obtained, the mean error is of the same order as in the identification positions and the maximum is below the maximum obtained in that case. It should be considered that the maximum values of standard deviation are obtained in the end spheres of the gauge, in more forced positions of the measurement arm. Given that we check the error values in one single test position of the gauge, better results in the arm behavior could be expected. However, it should be remembered that, for this ball bar location, a total of 6195 arm positions are evaluated, both from the point of view of point repeatability and error in distances based on the calculation of the mean point probed for each sphere. In measurement

tests taking 9 discrete points corresponding to the surface of the sphere and evaluating the six distances of the gauge, error values lower than the mean error corresponding to the evaluation by way of continuous capture on the test position are always obtained. For this reason, as the conclusion of the evaluation test, the importance of the data captured should be again emphasized. A high number of arm positions, different to those chosen for identification, should be searched in the way recommended in normalized evaluation test, in order to conclude with the acceptance of the identified model parameters. In this case, the number of arm positions considered for evaluation is high compared to those used in identification, obtaining values below the maximum error, meaning the arm behavior is verified in accordance with these maximum errors within the volume considered.

5.2. Repeatability error model

In order to check whether it is possible to generalize the repeatability error model outside of the calibration positions, the repeatability error is calculated and represented with respect to the mean in the test position probed, with the external data not previously used neither to identify the kinematic model parameters nor to obtain the Fourier series approximation. As a general rule, the consideration of functions with a bigger number of harmonics and without considering cross terms is a better approximation to the data captured for identification. However, in the case of evaluation of the error model on the test position, unwanted behavior of the error model can be observed when considering more harmonics, resulting in value error predictions around twice the real error in specific positions.

Taking the approximation function finally obtained for the error model, with two harmonics plus all their cross products, the evaluation on the test position is shown in Fig. 10. Again, for this figure, the real repeatability error occurring in the test position with the optimum kinematic parameters and the estimated error in each coordinate can be seen. The error model based on the function with two harmonics plus the cross products can be considered generalizable, behaving correctly in the evaluated positions. There are still areas without suitable approximation and even peaks which will make the maximum error bigger than the initial one both in coordinate Y and in Z, but the repeatability error descends considerably in mean value. Based on the tests carried out, it seems reasonable to use this approximation, which predicts great part of the errors observed in the validation position checked. Likewise, it does not seem necessary to increase the number of terms considered in the approximation.

Once obtained and validated the model corresponding to the repeatability error in the three coordinates, a dynamic correction of this error based on the model developed can be implemented. This implementation requires adding to the kinematic model of the AACMM the parametric correction functions chosen to predict the error. In this manner, the error model functions coefficients are stored separately, constituting an addition to the kinematic model function with error correction. Thus, Eq. (18)

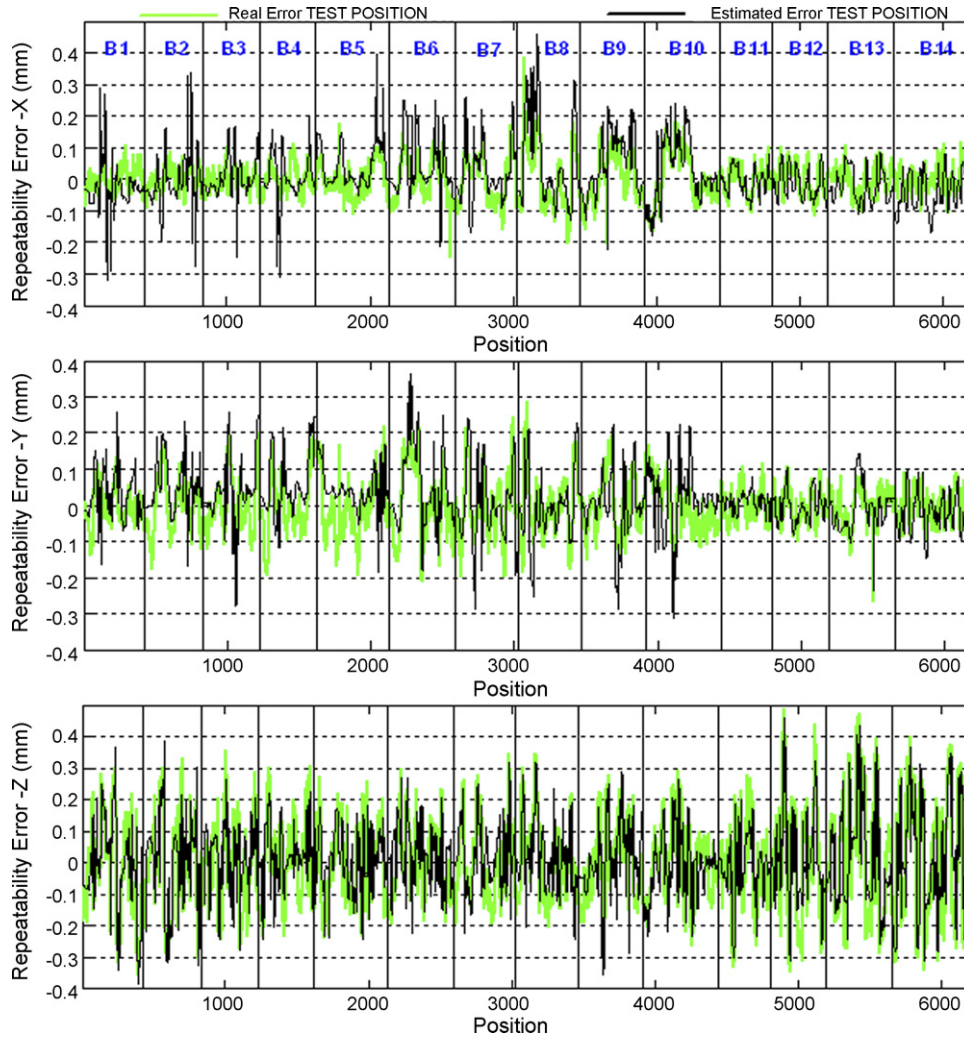


Fig. 10. Predicted and real repeatability error in ball bar test location. Approximation with 2 harmonics and cross products.

shows the obtaining of the measured final point, including the repeatability error correction.

$$\begin{aligned}
 X_{\text{GLOBAL-}i} &= X_{\text{MEAS-}i}(a_j, \alpha_j, d_j, \theta_{0j}, X_{\text{Probe}}, Y_{\text{Probe}}, Z_{\text{Probe}}, \theta_{\text{EncjMEAS}}) + f_{\varepsilon X}(\theta_{j\text{MEAS}}) \\
 Y_{\text{GLOBAL-}i} &= Y_{\text{MEAS-}i}(a_j, \alpha_j, d_j, \theta_{0j}, X_{\text{Probe}}, Y_{\text{Probe}}, Z_{\text{Probe}}, \theta_{\text{EncjMEAS}}) + f_{\varepsilon Y}(\theta_{j\text{MEAS}}) \\
 Z_{\text{GLOBAL-}i} &= Z_{\text{MEAS-}i}(a_j, \alpha_j, d_j, \theta_{0j}, X_{\text{Probe}}, Y_{\text{Probe}}, Z_{\text{Probe}}, \theta_{\text{EncjMEAS}}) + f_{\varepsilon Z}(\theta_{j\text{MEAS}})
 \end{aligned} \tag{18}$$

In Eq. (18), $X_{\text{GLOBAL-}i}$ represents coordinate X of the points obtained following the application of the kinematic model and correction, that is, the measured final point. $X_{\text{MED-}i}$ represents coordinate X of the point obtained by the kinematic model, which will be function of the model parameters, with j from 1 to 6, and of the joint angle values in current position i . Finally, $f_{\varepsilon X}(\theta_{j\text{MED}})$, which also depends on the angle values of each joint for the current position, contains the correction function obtained for coordinate X in the previous section, which is directly added to the point coordinate obtained by the kinematic model. This corrects the error, aiming to improve the point repeatability without affecting the accuracy of measurement in distances reached in calibration. In this way, the combined implementation of the correction mechanism and model will allow dynamically obtain-

ing, during the capture of points in measurement, the already corrected measured point values.

Given the approach of the method, the application of the correction on the measurements must have very little influence on the accuracy in distances of the arm, provided by the parameters identified. However, since this accuracy want to be preserved while the point repeatability error decreases, the application of the correction should reduce the mean deviation in each probed sphere. It is possible to evaluate the effect of the application of the correction implemented in Eq. (18) on the data of arm positions corresponding to the 14 spheres of the test position used in evaluation (Table 8), in order to check the characteristic metrological parameters in this case with respect to the values obtained for the test position without correction showed in Table 7. As

Table 8

Quality indicators for the identified set of model parameters over 14 spheres of ball bar test location (6195 AACMM positions) with repeatability error correction

Distance error (mm)		2 σ by Sphere (mm)	
Maximum	0.129830	Maximum	0.104020
Causing Pos.	POS.TEST	Causing Pos.	POS.TEST
Causing Dist.	D1-14	Causing Sph.	B12
Minimum	0.006726	Causing Coord.	Z
Causing Pos.	POS.TEST	Minimum	0.029521
Causing Dist.	D1-2	Causing Pos.	POS.TEST
Medium	0.073171	Causing Sph.	B11
		Causing Coord.	X
		Medium X	0.047699
		Medium Y	0.059570
		Medium Z	0.088127

can be seen in Table 8 there is slight worsening of the error value in distances when applying the correction model, although the change experienced both by the maximum value of this error and its mean value is negligible compared to the improvement experienced by the standard deviation values. This leaves to conclude that the error correction model is suitable to improve the mean repeatability of the arm for a given set of kinematic parameters.

6. Conclusions

This work has presented a procedure for kinematic calibration and error correction for AACMMs. Firstly, a new method for the capture of data for identification of AACMM kinematic parameters has been proposed. This method is based on the continuous capture of arm positions by directly probing the center of the spheres of a gauge ball bar by way of a kinematic coupling probe, unlike current methods which are based on the capture of identification data probing surface points of geometrical primitives of different gauge objects. This method is a quick and effective data capture technique in AACMM parameter identification procedures, allowing the coverage of a large number of arm positions in the measurement volume considered for calibration. Likewise, a new approximation to the AACMM kinematic parameter identification problem has been presented. This approximation is based on an objective function including terms of error in distances and terms of standard deviation which consider the influence of arm repeatability, given its capacity to probe the same point from different orientations. The objective functions used so far in robot arms and AACMMs only consider terms of error in distance between the point probed or reached and that acquired by an external gauge, without considering the probing effect from different orientations. As has been shown experimentally, the consideration solely of errors in distance in the objective function results in a set of parameters which obtain good error results in distances, but unacceptable repeatability errors for this type of instruments. This fact is even worse when considering low numbers of identification positions. The bibliography shows ambiguity regarding methods for kinematic parameter identification of robot arms with regards to the minimum number of identification positions necessary in order to

consider as generalizable the results obtained in a set of discrete identification positions. The method presented here allows making quick and accurate capture of nominal data for identification in order to consider the obtained accuracy and repeatability results as generalizable to all the measurement volume. The influence of the number of positions considered in the final error has been shown with experimental results. Finally, a dynamic correction model for repeatability error has been proposed. This model is based on the information captured during the identification procedure. By way of experimental tests it has been checked that this model does not affect accuracy in distances obtained with the parameters identified and improves arm repeatability by more than 50%. With regards to the correction models for robot trajectories, it was observed that the consideration of fewer points implies better approximation. When the number of points is increased, these approximations generally result in maximum residual values similar to the error value to approximate, meaning that, in the case of error correction models for robots, they are only applicable in specific zones and cannot be extrapolated to the entire work space. However, increasing the number of positions is recommendable both for parameter identification and for approximation of correction functions. Moreover, in some cases it has been observed that the implementation of correction models involves the variation of the kinematic parameter values. This assumption is erroneous, especially when this consideration is in addition to the fact that the positions captured for parameter identification are different to those captured in order to determine the approximated functions for correction. This causes variations in the error in distances which will be manifest when correcting the error, since the kinematic parameter values will be different, for which reason the calibration conditions of the robot arm are no longer fulfilled. Generally, the results obtained by these correction models offer good results in the tests carried out by numerous researchers, thanks to the end purpose the robot is destined to, but are unacceptable for AACMMs because of the aforementioned reasons. The procedures presented allow establishing the basis to deal with later thermal error models and to implement free software tools for their use in coordinate measurement with AACMMs.

The lack of normalization and the scant bibliographical resources existing in the field of AACMMs has been shown, partially alleviated through the appearance of the ASME B89.4.22-2004 standard, which establishes AACMM verification procedures which permits clearly comparing their metrological characteristics. The techniques presented in this work are easily generalizable to any current measurement arm.

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